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5/5/2026

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Poverty and socioeconomic factors and their relationship on educational attainment within Maryland

Introduction:

Educational attainment is considered to be one of the most important predictors when it comes to economic opportunities. However, despite this importance, educational outcomes are typically influenced by socioeconomic factors like poverty, unemployment, and household income. This project studies the relationship between poverty and educational attainment within Montgomery County Public Schools, and across Maryland counties. By analyzing socioeconomic indicators next to school-level educational outcomes, this study aims to better understand how economic disadvantages may impact graduation rates, attendance, and dropout patterns. Generally speaking, the purpose of this project is to identify patterns of education inequality, while also discovering how poverty related factors can vary geographically across Maryland. Understanding these relationships plays a key role in providing insight on where additional educational resources and policy change is most needed.

Data:

The data for this project comes from three datasets, two being the most prominent. All three of these datasets are focused on educational outcomes and socioeconomic indicators, along with other important factors like demographics. The information in these datasets is drawn from Maryland communities, either from Montgomery County exclusively, or county-wide across the state.

Dataset 1:

The first dataset comes from the Open Maryland Data portal. This dataset contains a multitude of important socioeconomic factors, all of which were helpful for identifying poverty across Maryland. The data is also spread across Maryland counties, rather than just being limited to Montgomery County. Some of the key variables include: poverty rate, median income, unemployment rate, and several others regarding educational attainment. This dataset played a role in a county-wide analysis of Maryland counties.

Dataset 2:

The second dataset used for this project comes from Datacensus.gov. Most of its information is sourced from the U.S census American Community Survey. Essentially, this dataset has observations based on poverty rates, income levels, and general demographic characteristics for Montgomery County. This dataset was used to provide poverty indicators and income rates for the project. Key variables include poverty rates, child poverty rate, median household income, and adults without high school diplomas.

Dataset 3:

The third dataset for this project is sourced directly from Montgomery County Public Schools. It contains details about high schools across Montgomery County. Some of the key variables would be: graduation rates, dropout rates, attendance rates, and FARM (Free and Reduced Meals) participation. During the course of the project, this dataset served as a primary source of information, especially for graduation rates.

Goals:

Throughout this project, I wanted to find out the relationship between socioeconomic factors, and their impact across Maryland counties. To proceed with this, I aimed to find the summary statistics within each dataset, as well as running an extensive exploratory data analysis across the sets. I also proceeded to run a correlation analysis for different variables, and created visualizations to better process the data at hand. In order to make predictions, I created a multi-linear regression model for graduation rates. To round it off, I also observed the disparity across schools in Montgomery county, as well as wealth gaps across different Maryland counties.

Tools and Methods:

Primarily for this project, I used R studio for my data cleaning, analysis, and visualization. Several R packages were used for this project, including tidyverse for data manipulation, and janitor for cleaning variable names. On top of that, ggplot2 and ggrepel were used for enhancing visualizations, while sf and tigris were used for geographic mapping. I also ran a number of statistical analyses on the data. This included exploratory data analysis, correlation analysis, simple linear regression, and multiple linear regression modeling. Geographic visualizations were also created to study poverty and income patterns across Maryland counties.

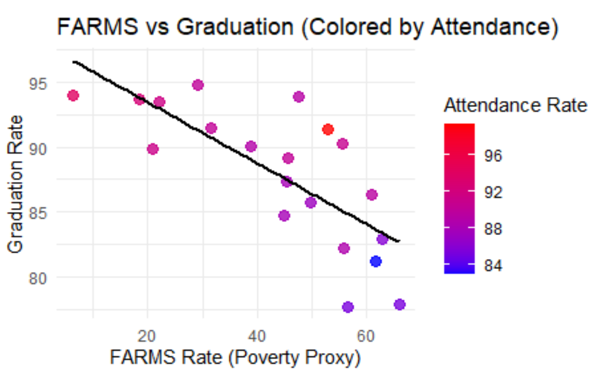
Data cleaning:

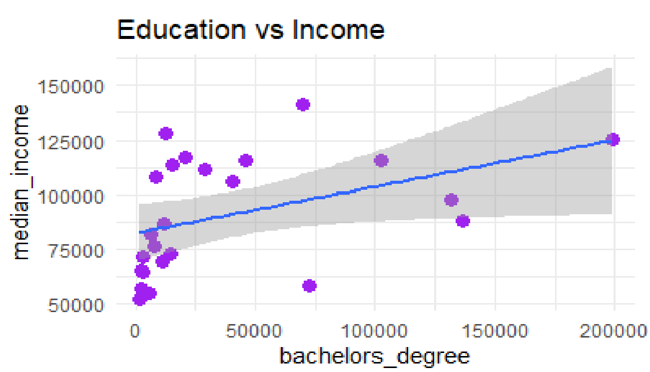
Once my data had been ingested, I had to commit to an extensive cleaning process. To begin, I created a new data frame for each dataset, labeling it as “dataset name_clean”. Through this, I renamed variables that were previously named oddly, to make them easier to work with. This was done with the mutate function, and the rename function as well. I also removed any missing values using the filter function, along with !is.na and the na.omit functions. Paired with that, some of my datasets were hard to use as their data was not formatted correctly. This resulted in numeric variables actually being classified as characters rather than numeric. I used mutate, and as.numeric to convert these variables to have actual values that could be graphed out. To make things easier, I also used the clean_names function under the Janitor package.

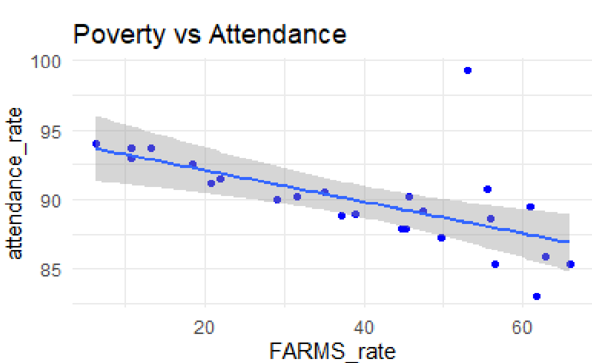
Summary Statistics:

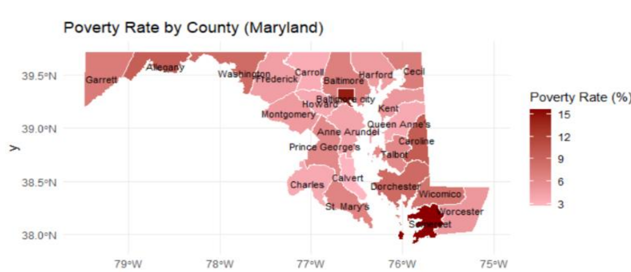
Overall, the summary statistics show that higher poverty indicators are associated with worse educational outcomes within Montgomery County, and across Maryland counties in general. Several percentage variables were initially stored as character strings, and required conversion in numeric format before statistical analysis could properly be conducted. Before proceeding, an important note to make is that FARMS rates represent food assistance in schools across Montgomery County. I used this as a poverty indicator when making observations. For the MCPS dataset, FARMS participation varied substantially across Montgomery County High Schools, ranging from 6.4% to 65.9%, with an average of 39.2%. This large spread suggests significant socioeconomic disparities between schools within the county.

Across counties, I was also able to discover valuable insights regarding poverty. County-level poverty rates across Maryland ranged from 2.8% to 15.6%, with an average poverty rate of 6.7%. This displays meaningful economic differences between counties in Maryland. There were also discoveries made regarding median county income. Median household income varied widely across Maryland counties, ranging from approximately 52,000 to over 140,000. This is a large disparity, and reflects substantial economic inequality distributed across the data.





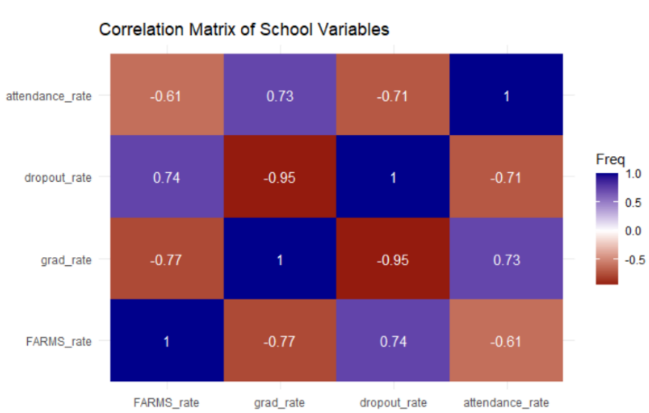


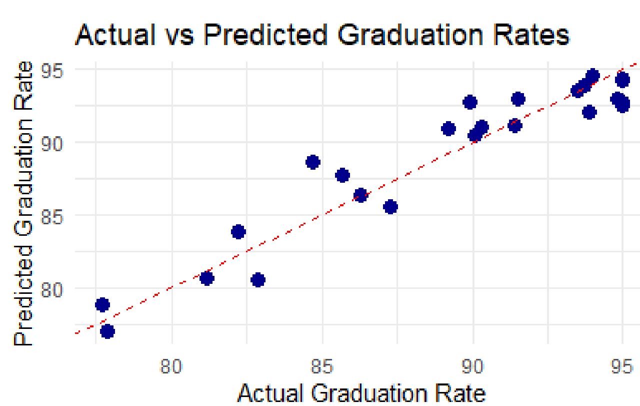


Final data product and revelations:

After summary statistics, I ran a few correlation analyses as well. The correlation between FARMS participation and graduation rate was -0.775, which shows a strong negative relationship between poverty and graduation outcomes. Schools with higher poverty levels generally had lower graduation rates. Poverty rates were strongly negatively correlated with median income (-0.791) and positively correlated with unemployment (0.659). These findings suggest that economic disadvantage is closely associated with both lower income and weaker labor market conditions across Maryland counties.

After the correlation analysis, I began working on my model. It is a linear regression model, made to predict graduation rates using factors like poverty, attendance, and dropout rates. The regression model explained approximately 91.2% of the variation in graduation rates ($R^2 = 0.912$), indicating a strong relationship between the selected predictors and educational outcomes. FARMS participation showed a negative relationship with graduation rates, although the effect was not statistically significant after accounting for attendance and dropout rates. This suggests that some effects of poverty may operate indirectly through other educational factors. Attendance rates showed a positive relationship with graduation outcomes, suggesting that schools with stronger attendance generally experienced higher graduation rates. Dropout rate was the strongest predictor in the model. A one-percentage-point increase in dropout rate was associated with approximately a 1.05 percentage point decrease in graduation rate, holding other variables constant. The overall regression model was statistically significant ($p < 0.001$), indicating that the predictors collectively explain graduation outcomes effectively.





Conclusion and final understandings:

Throughout this project, I examined the relationship between poverty, socioeconomic disadvantages, and educational attainment across Montgomery County and Maryland as a whole. Using county-level socioeconomic data alongside MCPS data, I was able to identify strong relationships between poverty indicators and educational outcomes. Schools with higher FARMS participation generally experienced lower graduation rates, while higher attendance rates were associated with stronger graduation outcomes. Across different Maryland counties, poverty was also closely connected to lower median income, and higher unemployment rates.

On top of that, the regression model further demonstrated that educational outcomes can be predicted reasonably well using variables related to poverty, attendance, and dropout rates. Although poverty alone was strongly associated with lower graduation outcomes, it appears that some of its effects also operate indirectly through attendance and dropout patterns. These findings reinforce the idea that educational inequality is closely tied to broader socioeconomic conditions. Overall, this project highlights the importance of targeting educational and economic support, especially within high poverty communities. By identifying where economic disadvantage and lower educational outcomes overlap, policy makers and school systems can better focus their resources towards schools and communities that need additional support. Education should be accessible.

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